

idTech4 ASE / LWO Tools

Blender 4.x / 5.x Static Mesh Import / Export Add-on

Version 1.0.0 · Author: Samson & Claude Sonnet

1. Overview

The idTech4 ASE / LWO Tools add-on imports and exports idTech4 static meshes in both formats the engine reads: .ase (ASCII Scene Export) and .lwo (LightWave Object, LWO2). It also reads the Doom 3 BFG compiled binary caches of each, .base and .blwo.

Every format quirk implemented here is verified against the Doom 3 GPL source rather than against a specification document, because the two disagree in places that matter — which layer of a .lwo is rendered, how a .ase's smoothing groups are treated, how a material name is derived from an *BITMAP path. Where the file and the engine disagree, the add-on lets you choose which one to follow.

Supported operations:

- Import .ase / .base — static mesh, one object per GEOMOBJECT
- Import .lwo / .blwo — static mesh, one object per layer
- Export .ase — with exact per-corner normals
- Export .lwo — with LWO smoothing groups, vertex maps and multi-layer output
- Auto-generate real, fully textured Blender materials on import (needs the companion "idTech4 Materials" add-on)
- Export Check panel — validate a model against what the engine actually accepts, and fix the fixable parts

This add-on is one of six that make up the idTech4 Blender toolchain. It installs and runs entirely on its own, but it also acts as a library for the others: the .map importer calls straight into this module to place the static models a map references, rather than carrying a second copy of the format code. When this add-on is absent, the .map importer greys out its own static-model option instead of failing.

2. Installation

1. Open Blender and navigate to Edit > Preferences > Add-ons.
2. Click Install... and select idTech4_ase_lwo_io.py.
3. Tick the checkbox next to "idTech4 .ase / .lwo Importer/Exporter".
4. Close Preferences. The add-on is now active.

After installation the following entries appear:

- File > Import > idTech4 ASE (.ase, .base) (static mesh)
- File > Import > idTech4 LWO (.lwo, .blwo) (static mesh)
- File > Export > idTech4 .ase (static mesh)
- File > Export > idTech4 .lwo (static mesh)
- 3D Viewport > N-Panel > idTech4 > Sources
- 3D Viewport > N-Panel > idTech4 > Export Check

3. Shared Settings

3.1 Source Directories

idTech 4 games do not ship their assets as loose files. Doom 3, Quake 4, Prey and The Dark Mod pack them into .pk4 archives; Doom 3 BFG packs them into .resources files. This add-on does not read inside either. It reads an ordinary directory tree on disk, and nothing else.

So before you set Base Directory, extract everything. A .pk4 is an ordinary zip archive — any zip tool will open one, and most will do it without renaming the file first. A .resources file is not a zip and needs a Doom 3 BFG extraction tool.

Extract every archive into one folder, preserving the paths stored inside them, so the result is the full idTech 4 directory structure:

```
<game>/
  base/                ← Base Directory points HERE
    def/               entityDef and model declarations
    dds/               precompressed textures
    guis/
    maps/              .map source files
    materials/         .mtr material declarations
    models/            .md5mesh, .md5anim, .ase, .lwo
    skins/             .skin declarations
    textures/          .tga / .png / .jpg source textures
    generated/         BFG only: .bimage, .bmd5mesh,
                      .bmd5anim, .base, .blwo caches
```

IMPORTANT: Base Directory is the "base" folder ITSELF — the one that CONTAINS materials/, models/, textures/ and the rest — not the game folder above it, and not one of the folders inside it. Every path in a .mtr, .def, .skin or .map is written relative to this folder, so

pointing one level too high or too low resolves nothing.

NOTE: A game ships several numbered archives (pak000.pk4, pak001.pk4, ... or game00.pk4, game01.pk4, ...) and the later ones patch the earlier ones. Extract them in ascending numeric order and let later files overwrite earlier ones — that is the same precedence the engine applies at run time. Extracting them in the wrong order silently leaves you with pre-patch assets.

For a mod or an expansion, extract its own archives into its own folder beside base — d3xp, darkmod, and so on — and set Mod Base Directory to that folder while leaving Base Directory on base. A mod supplies part of the tree and inherits the rest, and the add-ons search the two the same way the engine does.

NOTE: The extracted tree only has to contain what you actually intend to load. Sound, video and script archives can be left packed. The folders that matter are the ones in the diagram above.

Base Directory, Mod Base Directory and Materials Source are stored in a small JSON file under Blender's per-user config directory and shared by every idTech4 add-on. They are edited from the 3D Viewport N-Panel, idTech4 tab, Sources panel; the import dialogs show them read-only.

Setting	Required	Purpose
Base Directory	Yes	A standard Doom 3-style "base" folder — the one containing materials/, models/, textures/ and so on.
Mod Base Directory	No	An optional second tree searched BEFORE Base Directory, with Base Directory still searched behind it — the engine's own fs_game behaviour.
Materials Source	No	One .mtr file, or a folder of them. Blank falls back to the "materials" subfolder under Mod Base (if set) and then Base Directory.

If Generate Materials is ticked and no Base Directory is configured, the import stops and offers a one-off setup dialog: Select Now, Derive Automatically (just this time, or and save settings), or Skip. The dialog fires from inside the import, after the file and every option are already chosen, and only when it could actually accomplish something.

3.2 Scale and Rotation

Both importers and both exporters share the same Transforms box. Scaling is off by default, so nothing is scaled unless you choose otherwise.

Option	Type	Default	Description
Scale Mode	Enum	None	None, the idTech4 ↔ Blender unit preset (inches ↔ metres: ×0.0254 in, ×39.3701 out),

Option	Type	Default	Description
			or Custom Factor.
Scale Factor	Float	1.0	An arbitrary uniform multiplier, shown only under Custom Factor. Range 0.0001 – 10000.
Rotation	Enum	0°	A pure Z rotation: 0°, 90° (X to Y), -90° (Y to X), or 180°.

IMPORTANT: On EXPORT the Rotation dropdown lists the IMPORT presets and the exporter INVERTS whichever you pick. Select the same preset you used on import and the model goes back out in the orientation it came in with — do not select the inverse yourself. (This differs from the MD5 exporters, which apply the chosen rotation directly.)

On export, each object's full current Blender transform is baked in first, so moving, rotating or scaling an object in the viewport is reflected in the file, then the Rotation preset is undone, then the scale factor converts Blender units back to file units. Normals go through the same rotation — via an inverse-transpose, so non-uniform object scale does not skew them — but never through the scalar factor.

4. Importing

4.1 Import ASE

Use File > Import > idTech4 ASE (.ase, .base) (static mesh). The importer creates one collection named after the file containing one object per GEOMOBJECT in it. Object and mesh datablocks are named after the model piece itself (the *NODE_NAME), not a generated compound name.

Option	Type	Default	Description
Scale Mode / Factor	Enum / Float	None / 1.0	See Section 3.2.
Rotation	Enum	0°	See Section 3.2.
Shading	Enum	File Shading	How the imported mesh is shaded — see Section 4.3. Hidden behind an informational line for a compiled .base, which carries its own normals.
Generate Materials	Bool	On	Build real, fully textured materials from the .mtr tree instead of blank placeholders. Greyed out when the "idTech4 Materials" add-on is not installed and enabled.
Material Mode	Enum	Maximum	Which of the Materials add-on's fidelity modes to build with: Maximum, Good, Basic or Simple.

Option	Type	Default	Description
Parameters	<i>Enum</i>	Dynamic	What to do with time, parm0..11, global0..7, sound and table lookups: Dynamic, Baked or Skip.
Derive from model	<i>Bool</i>	Off	Ignore the three paths below and derive them for this import by walking up from the imported file's own directory looking for a "materials" folder.
Base / Mod Base / Materials	<i>Display</i>	—	Read-only mirrors of what this import will use. Hover for the full path; edit them in the Sources panel.

4.2 Import LWO

Use File > Import > idTech4 LWO (.lwo, .blwo) (static mesh). Same collection-per-file layout as the .ase importer, with one object per layer. The import is lossless by default: a Blender vertex index is the .lwo point index, so the file can be edited and written back unchanged.

Option	Type	Default	Description
Scale Mode / Factor / Rotation	<i>Enum / Float</i>	None / 1.0 / 0°	See Section 3.2.
Shading	<i>Enum</i>	File Shading	See Section 4.3.
Only import 1st layer	<i>Bool</i>	On	Import only the first layer, which is all the engine ever reads (idRenderModelStatic::ConvertLWOToModel Surfaces). Any additional layer is parsed but never rendered. Disable to import every layer, each as its own object. Hidden for a compiled .blwo.
Build Engine Render Mesh	<i>Bool</i>	Off	Build the mesh idTech4 actually RENDERS rather than the mesh the file describes — see the note below. Hidden for a compiled .blwo.
Generate Materials / Material Mode / Parameters	<i>Enum / Bool</i>	On / Maximum / Dynamic	Identical to the .ase importer — see Section 4.1.
Derive from model, Base / Mod Base / Materials	<i>Bool / Display</i>	Off / —	Identical to the .ase importer.

NOTE: Build Engine Render Mesh replicates the engine's own non-fastLoad load path: coincident points welded at r_slopVertex 0.01, texcoords welded at r_slopTexCoord 0.001, then one vertex per unique combination of position, texcoord, vertex colour and normal — so every UV seam and every hard edge becomes a real vertex split — degenerate triangles

dropped, and any vertex used by both texture polarities duplicated. The vertex count comes out HIGHER than the file's and the mesh no longer corresponds to it point-for-point. Use it to see exactly where the engine will split your model; do not use it as a source of truth about the file. Two caveats: Blender cannot hold two faces on one vertex set, so a triangle the weld collapses onto another is dropped here where the engine would keep it (the count is printed to the console); and a renderbump HIGH poly is loaded with fastLoad instead, which welds nothing at all.

4.3 Shading

Both importers offer the same four shading modes. The distinction matters because a file's own intent and the engine's rendering of it are not always the same thing.

Mode	What it does
File Shading (default)	Shade the way the file intends, which is what a non-idTech4 model wants. Identical to Engine Shading for every .lwo — LWO stores no normals, so its smoothing angle is both — and for any .ase that has *MESH_NORMALS. It differs only for an .ase without them, where this honours the *MESH_SMOOTHING groups the engine throws away.
Engine Shading	Shade the way idTech4 will actually render it. For a .lwo that is the file's SURF SMAN angle and PTAG SMGP groups (lwGetVertNormals); for an .ase it is *MESH_NORMALS, or — when the file has none — the normals the engine regenerates from geometry, which ignore *MESH_SMOOTHING entirely. With material generation on, a renderBump or unsmoothedTangents material is honoured too.
Smooth	No custom normals: every face smooth-shaded with Blender's own averaging. Matches no idTech4 asset — Blender weights by corner angle and crosses material boundaries, the engine does neither — but it is the only way to get an unconstrained mesh, since the two modes above write custom split normals that override Blender's own shading controls.
Flat	No custom normals, every face faceted. Same reasoning as Smooth.

NOTE: Vertex colours are imported to Blender's Face Corner colour attribute. That is the only Blender domain that can hold either format losslessly: .ase stores colour per face corner by construction (*MESH_CFACE indexes three entries per face) and .lwo VMAD is keyed by (polygon, point). The Vertex domain cannot express a colour that breaks at an edge, which both formats routinely carry.

The importer keeps quad and n-gon geometry as it finds it rather than silently triangulating or dropping it, which is useful for reviewing source art — but it means what you see can be MORE geometry than actually renders in-game. Any such object is named in the import's status message.

5. Exporting

IMPORTANT: Both exporters switch the active object to Object Mode for the duration and switch back afterwards. Blender does not write Edit Mode's working copy back to the mesh datablock until the mode is exited, and the exporters read that datablock — without the round trip, exporting from Edit Mode wrote the geometry as it was when Edit Mode was entered, with no error and no warning. Check Model and Prepare for Export carry the same guard.

5.1 Export Scope

Both export dialogs open with the same scope controls and a live manifest of exactly what is about to be written.

Option	Type	Default	Description
Scope	<i>Enum</i>	Collection	Collection: every mesh in the active object's collection — this is what an import produces, one collection per file, so it is the natural unit of "a model" and it ends the "did I select every piece?" failure. Selected Objects: only the selected meshes. Whole Scene: every mesh object in the scene.
Skip Hidden Objects	<i>Bool</i>	On	Leave out anything whose outliner EYE is closed. The monitor and camera icons are deliberately not consulted — one obvious control decides what reaches the file, and the manifest lists exactly what is being skipped, before you press Export.
Apply Modifiers	<i>Bool</i>	Off	Export each object's modifier-evaluated mesh instead of its raw geometry. Off by default so the mesh in the outliner is the mesh that gets written, an export round-trips this add-on's own import unchanged, and a Subdivision modifier left on a cage cannot silently multiply the exported triangle count.

The manifest box under the scope controls lists the object count, the triangle count after triangulation, the first few object names, how many non-triangle polygons are present and whether they will be triangulated, and — in red — anything hidden that is being skipped. It refreshes on every redraw and counts polygons only; the expensive checks live behind the Check Model button (Section 6).

5.2 Export ASE

Use File > Export > idTech4 .ase (static mesh). The filename is pre-filled from the object about to be exported when there is one obvious candidate. .ase is triangle-only by construction, so everything in scope is triangulated on the way out.

Option	Type	Default	Description
Export Normals/Smoothing	<i>Bool</i>	On	Write each face corner's exact current Blender shading normal (flat, smooth, custom split, Smooth-by-Angle — whatever the viewport shows) as *MESH_NORMALS data. Real idTech4 uses this directly for any material without a renderBump command. Disable to omit *MESH_NORMALS entirely and let the engine generate its own, always fully smoothed, normals instead.
Export Vertex Colors (Face Corner)	<i>Bool</i>	On	Export Blender's Face-Corner colour attributes.
Export Vertex Colors (Vertex)	<i>Bool</i>	Off	Convert Vertex-domain colour attributes to Face Corner and export them too.

NOTE: .ase is the format to use when exact per-corner normal fidelity matters. It is the only one of the two that can carry an arbitrary custom split normal.

5.3 Export LWO

Use File > Export > idTech4 .lwo (static mesh). A .lwo never stores per-vertex normals; the engine always recomputes them at load time from the surface's SMAN smoothing angle and the polygons' SMGP smoothing groups.

Option	Type	Default	Description
Merge Into Single Layer	<i>Bool</i>	On	Combine every exported object into ONE LWO2 layer, matching the engine, which only ever reads a .lwo's first layer. Disable to write one file still, but with one layer per object — good for a faithful round trip or for further authoring in LightWave, but be aware the engine will render only the first layer and silently ignore every other one.
Smoothing Groups (SURF SMAN / PTAG SMGP)	<i>Bool</i>	On	Reproduce each triangle's flat/smooth shading through LWO's own smoothing-angle and smoothing-group mechanism — the only mechanism the engine ever uses to shade a .lwo. Exactly reproduces a flat/smooth per-face toggle; cannot reproduce an arbitrary custom split normal or a numeric Smooth-by-Angle threshold. Disable to force every face fully faceted (SMAN 0).
Preserve File Smoothing	<i>Bool</i>	On	Write back the SMAN angle and SMGP groups the model was imported with, so the engine recomputes exactly the shading it had. Turn it off to publish your Blender flat/smooth

Option	Type	Default	Description
			edits instead. Ignored for a mesh with no imported smoothing data, which always uses the flat/smooth flags.
Export Vertex Colors (Face Corner)	<i>Bool</i>	On	Export Blender's Face-Corner colour attributes as VMAD.
Export Vertex Colors (Vertex)	<i>Bool</i>	Off	Convert Vertex-domain colour attributes to Face Corner and export them too.
Triangulate	<i>Bool</i>	On	Split every polygon into triangles. Required for idTech4: the engine renders only already-triangulated polygons and drops anything else outright. Turn it off to write quads and n-gons as-is for a faithful round trip of a non-idTech4 model — the result will not render in-game.
All Vertex Maps	<i>Bool</i>	Off	Write every UV map, corner colour attribute, vertex group, selection set and shape key the mesh carries, plus polygon part tags. The engine reads none of this beyond one UV and one colour, so it is dead weight in-game and matters only for a faithful non-idTech4 round trip. Off writes just the active UV and colour map.
Weld Coincident Points	<i>Bool</i>	Off	Merge points that round to the same position, across every exported object, into one shared PNTS entry — the way a hand-modelled multi-part LWO layer would be built. UV and colour are unaffected (both are written per (polygon, point)), but geometry is not: welding can collapse a triangle into a degenerate or a duplicate and nothing detects that. Off by default because the engine never welds a .lwo on load, so leaving it off round-trips this add-on's own lossless import unchanged.

6. The Export Check Panel

3D Viewport > N-Panel > idTech4 tab > Export Check. Two buttons and a snapshot of the last run. The panel sits directly under the shared Sources panel.

6.1 Controls

Option	Type	Default	Description
Scope	<i>Enum</i>	Collection	Which objects to look at — the same three choices, resolved the same way, as the

Option	Type	Default	Description
			export dialogs use.
Target	<i>Enum</i>	.lwo and .ase	Which format's constraints to report: both, .lwo only, or .ase only.
Skip Hidden Objects	<i>Bool</i>	On	Leave out anything whose outliner EYE is closed.

6.2 Check Model for idTech4

Runs the engine's own acceptance rules over everything in scope and writes a full report to the system console. It changes nothing: meshes are evaluated the same way the exporter evaluates them and nothing in the scene is touched. Findings are grouped into three tiers.

Tier	Meaning
Needs a decision from you	Things no automated fix can settle: a polygon over 1023 vertices, a missing UV map, a degenerate UV triangle, a material identified as not idTech4-authored, a material path with a dot in a folder name, a material name the engine cannot resolve.
Prepare for Export can fix	Non-triangle polygons and zero-area faces — press the second button.
The exporter handles, for information	Things that are written correctly anyway but are worth knowing about, such as extra UV maps that the engine will flatten into one.

The box above the buttons redraws what the last run found: object and triangle counts, the object names, any non-triangle polygons, whether the output would be merged into one .lwo layer, and anything hidden that was skipped. It is a snapshot, not a live count, and it flags itself stale when the Scope, Target or Skip Hidden controls above it no longer match the run it is describing.

NOTE: An object can be excluded from every scope explicitly by setting a custom property named `idtech4_exclude` on it. Objects skipped that way are listed separately in the report.

6.3 Prepare for Export

Applies the two fixes the checker classes as fixable. Its three settings sit under the button and are read at run time, so they are chosen before the run rather than discovered afterwards in Adjust Last Operation. The run is a normal undoable operation either way.

Option	Type	Default	Description
Work on a Copy	<i>Bool</i>	Off	Duplicate everything in scope into a new "<collection>_idtech4" collection and prepare that, leaving the authoring meshes untouched, so the export-ready model sits

Option	Type	Default	Description
			beside the cage you keep modelling on. Off prepares the originals in place.
Triangulate	<i>Bool</i>	On	Split every polygon into triangles. The engine drops anything that is not one.
Remove Zero-Area Faces	<i>Bool</i>	On	Delete faces the engine would discard at load anyway (R_RemoveDegenerateTriangles). Neither exporter does this, so it is the one fix only this button applies.

7. Format Notes

7.1 Which Layer the Engine Reads

Verified against `idRenderModelStatic::ConvertLWOToModelSurfaces`: the engine only ever reads the FIRST layer of a .lwo. Any additional layer is parsed and then never rendered. That is why "Only import 1st layer" and "Merge Into Single Layer" both default to on — they keep Blender's view of the model and the engine's view of it the same. Turn them off only when you are deliberately working with a multi-part LightWave object rather than a game asset.

7.2 Material Names

The exported material or surface name IS the in-game material path. A .lwo TAGS/SURF name goes straight to `declManager->FindMaterial()`; an .ase carries it in *BITMAP, which the engine puts through `OSPathToRelativePath` first. These are different kinds of string, so the exporters verify what they write rather than assuming a name round-trips.

Material names are matched the way the engine matches them: backslashes converted to slashes, lowercased, and truncated at the last dot. A surface spelled with either separator therefore resolves to the same decl. This is why Check Model flags a material path with a dot in a folder name — the truncation would cut the path short.

7.3 Smoothing, and What Each Format Can Carry

An .ase can carry an arbitrary per-corner normal, so a Blender mesh's shading survives an .ase round trip exactly. A .lwo cannot: SMAN plus SMGP reproduce a per-face flat/smooth toggle and nothing finer. If exact per-corner fidelity matters more than authoring in LightWave, export .ase.

Smoothing-group ids are assigned by greedy colouring of the flat groups, which keeps the id count inside what the format allows on models where a naive per-group id would overflow.

7.4 Binary Caches

Doom 3 BFG ships compiled caches of its static models: .base for .ase and .blwo for .lwo, under generated/. Both are read by the matching importer. They carry finished normals, so the

Shading control shows an informational line instead of the geometry options for them. There is no binary exporter — export always writes the text/IFF format.

8. Tips and Troubleshooting

Nothing in Scope

The export dialog refuses with "Nothing in scope" when the manifest is empty. With Collection scope this usually means the active object is not in the collection you meant; with Skip Hidden Objects on it can mean every mesh in scope is hidden, which the error message says explicitly.

The Model Renders With Holes In-Game

Almost always non-triangle polygons. The engine drops them outright with "make sure you triplet it down". Run Check Model, then Prepare for Export, or simply leave Triangulate on in the .lwo exporter (.ase is always triangulated).

The Model Renders Faceted In-Game But Smooth in Blender

For .ase, check that Export Normals/Smoothing was on — without *MESH_NORMALS the engine regenerates fully smoothed normals of its own. For .lwo, remember the format cannot carry custom split normals at all; the engine recomputes shading from SMAN and SMGP, so a Smooth-by-Angle threshold or a hand-edited normal will not survive. Export .ase instead.

Only Part of the Model Appears In-Game

A multi-layer .lwo. The engine reads the first layer only. Re-export with Merge Into Single Layer on.

A Re-Export Does Not Match the Original File

Leave Weld Coincident Points off and Apply Modifiers off. With both off, a .lwo import/export round trip is lossless — a Blender vertex index is the .lwo point index.

Nothing Resolves At All

If every lookup fails — no materials found, no models found, no .def files found — the usual cause is that the game's assets are still inside their .pk4 or .resources archives. These add-ons cannot read into an archive; everything has to be extracted to a real directory tree first. The second most common cause is Base Directory pointing at the game folder rather than at the base folder inside it: it must be the folder that directly contains materials/, models/ and textures/.

Materials Look Flat White

The material decls could not be found. Check that Base Directory (or Mod Base) points at a tree containing the right .mtr files, and read the build report the import writes to the Text Editor — it names every material that fell back to a placeholder and why. For The Dark Mod in particular, note that almost all textures live only under dds/.

Custom Normals and Bumpmapping

Custom split normals do not fix the tangent basis. A model whose faces are flat-shaded gives per-face tangents, and a bumpmapped material then looks faceted no matter what the normals say. Smooth fans are what a correct tangent basis needs.

idTech4 ASE / LWO Tools — end of documentation